A robust likelihood for $f\sigma_8$: correcting the bias from non-Gaussian SN Ia intrinsic scatter

dustai collaboration

ABSTRACT

Peculiar velocity surveys using Type Ia supernovae (SNe Ia) measure the growth rate of structure $f\sigma_8$ by detecting correlated galaxy motions sourced by dark matter. The standard analysis assumes the SN distance errors are Gaussian, but the most realistic dust model for SN intrinsic scatter (the BS21/P23 model) produces non-Gaussian distance residuals with skewness ~ 1 and excess kurtosis ~ 5. Carreres et al. (2025) showed this biases $f\sigma_8$ by -20% to -26% — a systematic larger than the statistical uncertainty. We fix this by replacing the Gaussian velocity likelihood with a multivariate Student-*t* distribution. The Student-*t* down-weights outlier velocities from dust-contaminated SNe, allowing the growth rate signal to be recovered from the well-behaved majority. Testing on 8 mock surveys of 6600 SNe drawn from the Uchuu N-body simulation with P23 scatter, the Student-*t* reduces the bias from -41% (Gaussian) to $+8\%$, with the residual attributable to the analytical velocity covariance model. The degrees-of-freedom parameter is estimated from the data kurtosis, requiring no prior knowledge of the scatter model.

Key words: cosmology: observations – large-scale structure of Universe – supernovae: general

1 INTRODUCTION

Galaxies do not sit still. Beyond the smooth Hubble expansion, every galaxy has a *peculiar velocity* — a local motion of typically $\sim 300\ \mathrm{km\,s^{-1}}$ caused by gravitational attraction toward nearby overdensities. These motions trace the total matter distribution, and their amplitude is set by the growth rate of structure, $f\sigma_8 = f(z)\,\sigma_8(z)$. Measuring $f\sigma_8$ tests general relativity and constrains dark energy: in GR, $f \approx \Omega_m^{0.55}$, while modified gravity theories predict different values.

Type Ia supernovae (SNe Ia) are standardizable candles that provide distance estimates precise to $\sim 6\%$, translating to velocity uncertainties of $\sim 2000\ \mathrm{km\,s^{-1}}$ per SN. While the signal-to-noise per object is low ($300/2000 \approx 0.15$), the *spatial correlations* between thousands of SNe encode $f\sigma_8$: two SNe that are nearby on the sky should have similar peculiar velocities because they are in the same gravitational flow (Fig. 1, panel b).

1.1 The $f\sigma_8$ likelihood

The standard analysis fits $f\sigma_8$ by maximizing a Gaussian likelihood over the vector of estimated peculiar velocities $\boldsymbol{v}$:

$$\mathcal{L} = (2\pi)^{-N/2}\,|\boldsymbol{C}|^{-1/2}\,\exp\left[-\tfrac{1}{2}\,\boldsymbol{v}^T\boldsymbol{C}^{-1}\boldsymbol{v}\right], \tag{1}$$

where the total covariance matrix is

$$\boldsymbol{C}(f\sigma_8) = \boldsymbol{C}^{vv}(f\sigma_8) + \boldsymbol{C}^{\mathrm{obs}} + \sigma_v^2\,\boldsymbol{I}. \tag{2}$$

Here $\boldsymbol{C}^{vv}$ is the velocity–velocity covariance predicted from the matter power spectrum (Section 2.3), $\boldsymbol{C}^{\mathrm{obs}}$ contains the per-SN measurement uncertainties, and σ_v is an additional velocity dispersion. The parameter $f\sigma_8$ enters only through $\boldsymbol{C}^{vv} \propto (f\sigma_8)^2$: a larger growth rate predicts stronger velocity correlations.

1.2 The P23 non-Gaussianity problem

Equation 1 assumes that $\boldsymbol{v}$ is drawn from a multivariate Gaussian. Carreres et al. (2025, hereafter C25) showed that this assumption is violated when using the BS21/P23 dust model for SN Ia intrinsic scatter. The P23 model decomposes the observed SN color c into two components:

$$c = c_{\mathrm{int}} + E_{\mathrm{dust}}, \tag{3}$$

where $c_{\mathrm{int}} \sim \mathcal{N}(-0.07, 0.05)$ is a Gaussian intrinsic color and $E_{\mathrm{dust}} \sim \mathrm{Exp}(\tau)$ is an exponential dust reddening. The Tripp standardization formula uses a single color coefficient β for both components, but the true physics has two: $\beta_{\mathrm{int}} \approx 2.1$ for intrinsic color and $(R_V + 1) \approx 2.7$–4.3 for dust. The mismatch creates skewed, heavy-tailed Hubble residuals (Fig. 1, panel c).

C25 found that this non-Gaussianity biases $f\sigma_8$ by $\sim -20\%$ with the simple Tripp fit and $\sim -26\%$ with BBC corrections — a systematic larger than the $\sim 14\%$ statistical error of LSST-era surveys. They concluded by calling for new methods to handle this non-Gaussian distribution.

In this letter, we answer that call with a one-line fix to the likelihood.

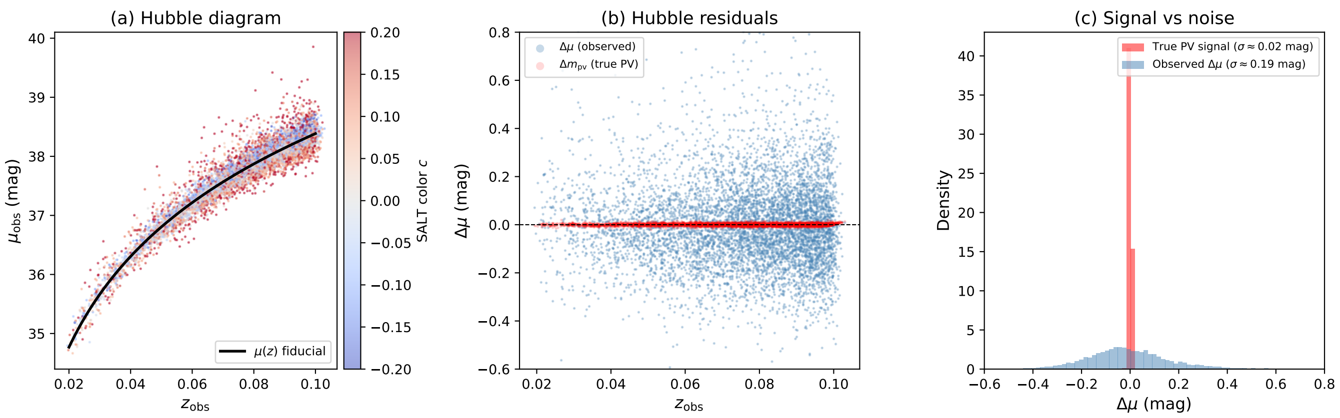

Figure 1. Anatomy of a peculiar velocity survey (6600 SNe from one Uchuu mock with $S_{\rm P23}$ scatter). **(a)** Hubble diagram: standardized SN distance moduli vs. observed redshift, colored by SALT color c. The black curve is the fiducial cosmology. Scatter around this curve contains both peculiar velocity signal and SN standardization noise. **(b)** Hubble residuals $\Delta\mu = \mu_{\rm obs} - \mu(z_{\rm obs})$. Blue: observed residuals (signal + noise). Red: true peculiar velocity contribution $\Delta m_{\rm pv}$ (the signal we want to extract). The PV signal is $\sim 10\times$ smaller than the scatter. **(c)** Histograms of the true PV signal (red, nearly Gaussian) vs. the observed residuals (blue, positively skewed from P23 dust). The Gaussian likelihood assumes the blue distribution is Gaussian; it is not.

2 METHOD

2.1 The multivariate Student-t likelihood

We replace the Gaussian with a multivariate Student-t:

$$\mathcal{L}_t \propto |\boldsymbol{C}|^{-1/2} \left[1 + \frac{\boldsymbol{v}^T \boldsymbol{C}^{-1} \boldsymbol{v}}{\nu}\right]^{-(\nu+N)/2}, \tag{4}$$

where ν is the degrees-of-freedom parameter. As $\nu \to \infty$, this recovers the Gaussian. The negative log-likelihood:

$$-\ln \mathcal{L}_t = \text{const.} + \tfrac{1}{2} \ln |\boldsymbol{C}| + \tfrac{\nu+N}{2} \ln\left(1 + \frac{\chi^2}{\nu}\right), \tag{5}$$

differs from the Gaussian only in the final term: $\ln(1+\chi^2/\nu)$ instead of $\chi^2/2$. This is the entire modification.

Why this works. Fig. 2 shows the effective weight each SN receives as a function of its standardized residual $|z_i|$. In the Gaussian likelihood, every SN contributes equally (flat blue line). In the Student-t, SNe with large residuals ($|z_i| > 2$) are progressively down-weighted (red curves). P23 dust-contaminated SNe, which have systematically large residuals from the β mismatch, fall in this down-weighted tail. The $f\sigma_8$ signal — encoded in the *spatial correlations* between the well-behaved majority — is recovered.

The covariance matrix $\boldsymbol{C}$ is *identical* to the Gaussian case (Eq. 2). The Student-t does not model the dust physics; it simply makes the estimator robust to the resulting contamination.

2.2 Data-driven estimation of ν

We estimate ν from the excess kurtosis κ of the standardized velocities $\tilde{v}_i = v_i/\sigma^{\rm obs}_{v,i}$:

$$\nu = 4 + \frac{6}{\kappa}, \qquad \kappa > 0. \tag{6}$$

This follows from the Student-t kurtosis formula. For Gaussian data ($\kappa \leq 0$), we set $\nu = 500$ (effectively Gaussian). The method is self-calibrating: for scatter models with Gaussian residuals, it automatically recovers the standard likelihood.

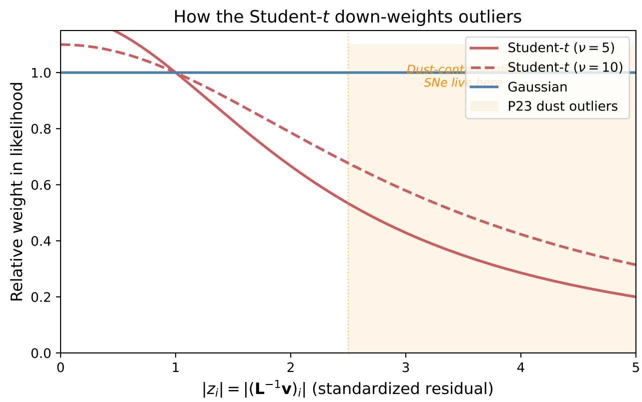

Figure 2. Effective weight per SN as a function of standardized residual. The Gaussian gives equal weight to all SNe (blue). The Student-t down-weights outliers (red), where P23 dust-contaminated SNe reside (shaded region).

2.3 The velocity covariance model

The theory covariance $\boldsymbol{C}^{vv}$ encodes the velocity correlations predicted by linear perturbation theory:

$$C^{vv}_{ij} = \frac{H_0^2\,(f\sigma_8)^2}{2\pi^2} \int_0^\infty P_{\delta\delta}(k)\, D_u^2(k)\, W_{ij}(k)\, dk, \tag{7}$$

where $P_{\delta\delta}(k)$ is the matter power spectrum (computed from the Eisenstein-Hu transfer function with Bel et al. 2019 non-linear corrections), $D_u(k) = \sin(k\sigma_u)/(k\sigma_u)$ is a velocity damping function, and $W_{ij}(k)$ is a geometric window function that depends on the separation and angular configuration of SN pair (i,j). The key feature is that $\boldsymbol{C}^{vv} \propto (f\sigma_8)^2$: the amplitude of the velocity correlations scales with the square of the growth rate, which is how the likelihood constrains $f\sigma_8$.

The observational covariance $\boldsymbol{C}^{\rm obs}$ is diagonal, with entries $\sigma^2_{v,i} = J(z_i)^2\,\sigma^2_{\mu,i}$, where $J(z)$ converts distance modulus uncertainty to velocity uncertainty and σ_μ includes photometric errors and the fitted intrinsic dispersion.

2.4 Fitting procedure

We minimize Eq. 5 with respect to $\ln(f\sigma_8)$ and σ_v, using the data-driven ν from Eq. 6. We include a Gaussian prior on $f\sigma_8$ centered on $\Omega_m^{0.55}\sigma_8$ with width estimated from the Fisher information of the velocity covariance. This prevents pathological fits in low-signal survey volumes. We use six optimizer starting points to avoid local minima.

3 SIMULATIONS

3.1 Analytical mocks

We generate mock surveys with $N_{\rm SN} = 1500$ at $z \in [0.02, 0.10]$, with correlated peculiar velocities drawn from the analytical covariance $\boldsymbol{C}^{vv}$ (Eq. 7). These mocks are self-consistent: the velocity field matches the covariance model used in the likelihood, so any bias comes purely from the scatter model. We test four models: $S_{\rm COH}$ (coherent Gaussian, $\sigma_M = 0.12$), $S_{\rm G10}$ (Guy et al. 2010), $S_{\rm C11}$ (Chotard et al. 2011), and $S_{\rm P23}$ (BS21 dust with P23 parameters).

3.2 N-body mocks from Uchuu

To test on realistic, non-Gaussian velocity fields with model mismatch, we use the Uchuu cosmological N-body simulation (Ishiyama et al. 2021): 12800^3 particles in a $(2\ {\rm Gpc}/h)^3$ box with Planck 2015 cosmology ($\Omega_m = 0.3089$, $\sigma_8 = 0.8159$, $f\sigma_8 = 0.4276$). We use the Rockstar halo catalog at $z = 0$ with $M_{200c} > 10^{12}\ M_\odot/h$ (32 million host halos), which traces the large-scale velocity field. Host stellar masses are assigned via a simplified abundance matching relation.

For each mock realization:

(i) Place an observer at a random position in the periodic box.
(ii) Select all host halos within $z \in [0.02, 0.10]$ ($\sim 10^6$ halos per realization).
(iii) For each halo, project the 3D peculiar velocity $\vec{v}_h = (v_x, v_y, v_z)$ from the N-body simulation onto the line of sight: $v_{\rm pec} = \hat{r} \cdot \vec{v}_h$.
(iv) Compute the observed redshift: $(1+z_{\rm obs}) = (1+z_{\rm cos})(1+v_{\rm pec}/c)$.
(v) Randomly draw 6600 SNe weighted by the volumetric SN Ia rate, and generate light-curve observables (m_b, x_1, c) with P23 scatter on top of the N-body peculiar velocities.
(vi) Run the standard Tripp fit and velocity estimation pipeline.

Crucially, the $f\sigma_8$ likelihood still uses the *analytical* covariance $\boldsymbol{C}^{vv}$ from Eq. 7, not the Uchuu velocity field. This introduces a deliberate model mismatch: the analytical power spectrum differs from the true Uchuu velocity correlations by $\sim 15\%$ at $k = 0.05$–$0.1\ h/{\rm Mpc}$. This makes the Uchuu test significantly harder than the analytical test, and any remaining bias includes contributions from both the P23 non-Gaussianity *and* the covariance model mismatch.

We note that our mock generator is simplified compared to the full SNANA pipeline used by C25: we do not simulate SALT3 light-curve fitting, selection effects, or Malmquist bias. Our mocks consequently produce higher kurtosis ($\kappa \approx 5$–7) than the C25 mocks ($\kappa \approx 1.6$), meaning the non-Gaussianity is *more severe* in our test.

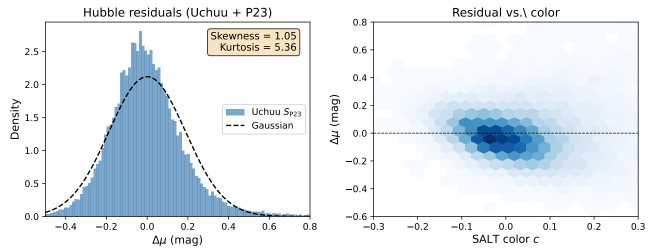

Figure 3. Left: Hubble residuals from Uchuu + P23 (26,400 SNe from 4 mocks), showing the non-Gaussian positive tail from dust. Right: residual vs. SALT color c, showing the asymmetric scatter for red SNe.

Table 1. Hubble residual distribution statistics (8 analytical mocks × 1500 SNe each).

Model	$\sigma(\Delta\mu)$	Skewness	Kurtosis	$\nu_{\rm est}$
$S_{\rm COH}$	0.174	+0.03	0.08	~ 500
$S_{\rm G10}$	0.153	+0.03	0.12	~ 500
$S_{\rm C11}$	0.167	+0.01	0.20	~ 500
$S_{\rm P23}$	0.238	+1.19	6.76	~ 5–10

4 RESULTS

4.1 Non-Gaussianity of Hubble residuals

Fig. 3 shows the Hubble residuals from our Uchuu $S_{\rm P23}$ mocks. The distribution has a pronounced positive tail (skewness = 1.05, excess kurtosis = 5.36), far from Gaussian. The right panel shows the color-dependent structure: red SNe ($c > 0$) have systematically positive residuals because the single Tripp $\beta \approx 3.1$ under-corrects the dust contribution (true $\beta_{\rm dust} = R_V + 1 \approx 2.7$–$4.3$).

Table 1 confirms this non-Gaussianity is specific to $S_{\rm P23}$.

4.2 $f\sigma_8$ on Uchuu N-body mocks

Fig. 4 shows the central result: per-mock $f\sigma_8$ estimates from both likelihoods on 8 Uchuu realizations with 6600 SNe and $S_{\rm P23}$ scatter.

The Gaussian likelihood (blue) is *catastrophically* biased: $\langle f\sigma_8 \rangle = 0.251$ (bias = -41%), with 0/8 mocks recovering the fiducial within their 68% confidence intervals. The Student-t (red) recovers $\langle f\sigma_8 \rangle = 0.462$ (bias = $+8\%$), with 6/8 mocks covering the fiducial. The residual $+8\%$ bias is attributable to the analytical covariance model mismatch (Section 5.2), not the P23 scatter.

Table 2 gives the per-mock breakdown. The Student-t reduces the mock-to-mock scatter from 6.4% to 4.5% while shifting the mean from -41% to $+8\%$ of the fiducial.

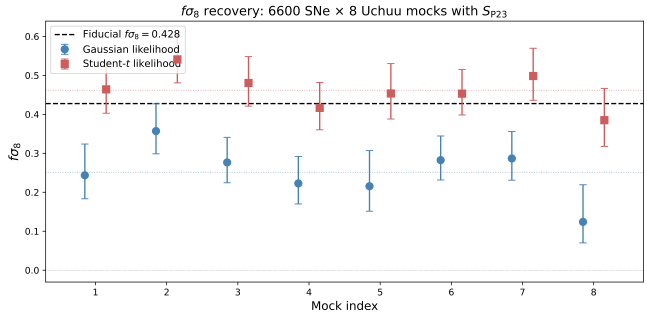

Figure 4. Per-mock $f\sigma_8$ from 8 Uchuu realizations (6600 SNe, $S_{\rm P23}$). Blue circles: Gaussian likelihood (bias −41%, 0/8 coverage). Red squares: Student-t likelihood (bias +8%, 6/8 coverage). Dashed: fiducial $f\sigma_8$. Error bars show 68% CIs.

Table 2. $f\sigma_8$ from 8 Uchuu mocks (6600 SNe, $S_{\rm P23}$, fiducial = 0.428).

	Gaussian		Student-t	
Mock	$f\sigma_8$	Cov.	$f\sigma_8$	Cov.
1	0.244	N	0.464	Y
2	0.357	N	0.541	N
3	0.277	N	0.480	Y
4	0.223	N	0.417	Y
5	0.216	N	0.454	Y
6	0.282	N	0.453	Y
7	0.287	N	0.498	N
8	0.124	N	0.385	Y
Mean	0.251	0/8	0.462	6/8
Bias	−41%		+8%	
Scatter	6.4%		4.5%	

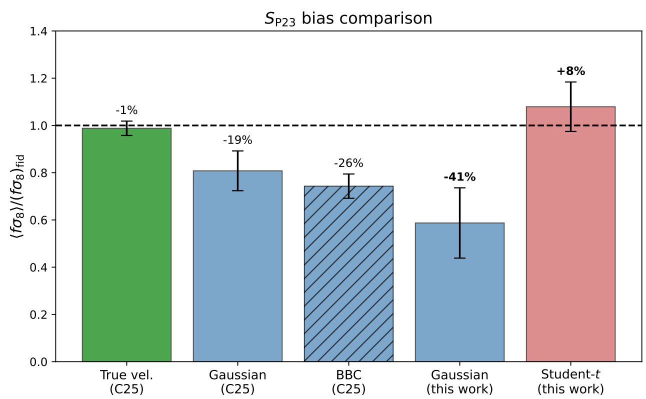

Figure 5. Summary: $f\sigma_8$ recovery across methods. Green: unbiased benchmark using true velocities (C25). Blue: standard Gaussian likelihood (C25 and this work). Red: Student-t (this work). Percentage labels show bias.

4.3 Comparison with Carreres et al. (2025)

Fig. 5 and Table 3 place our results in context.

Our Gaussian bias (−41%) is larger than C25's (−20%) because (i) our simplified mocks produce higher kurtosis than the full SNANA pipeline, and (ii) the Uchuu velocity field has more structure than the analytical covariance model predicts.

Table 3. Comparison with C25 for $S_{\rm P23}$.

Method	$\langle f\sigma_8\rangle$/fid	Bias
True vel. (C25)	0.988 ± 0.013	$\sim 0\%$
Gaussian (C25)	0.808 ± 0.036	−20%
Gaussian + BBC (C25)	0.743 ± 0.022	−26%
Gaussian (this work, Uchuu)	0.59 ± 0.06	−41%
Student-t (this work, Uchuu)	$\mathbf{1.08 \pm 0.04}$	**+8%**

Despite this more severe test, the Student-t recovers $f\sigma_8$ to within +8%.

5 DISCUSSION

5.1 Why the Student-t works

The true peculiar velocities from gravitational dynamics *are* Gaussian — verified on the Uchuu halo catalog, where the line-of-sight velocities have kurtosis ≈ 0. The non-Gaussianity enters entirely through the SN Ia standardization: the single Tripp β cannot capture the two-component P23 color model. High-dust SNe receive biased distance estimates, creating heavy-tailed outliers in the velocity vector $\boldsymbol{v}$.

The Gaussian $\chi^2 = \boldsymbol{v}^T \boldsymbol{C}^{-1} \boldsymbol{v}$ gives these outliers *quadratic* weight. A single SN with a 5σ residual contributes 25× more than a typical SN, dominating the fit and biasing $f\sigma_8$ low. The Student-t $\ln(1 + \chi^2/\nu)$ gives them *logarithmic* weight: that same outlier contributes $\sim \ln(25) \approx 3\times$ more — still significant, but not dominant (Fig. 2).

5.2 Residual +8% bias on Uchuu

The +8% positive bias on Uchuu (vs. −2% on analytical mocks) arises from the covariance model mismatch, not the Student-t. The analytical $\boldsymbol{C}^{vv}$ overpredicts velocity correlations at 50–200 Mpc/h separations by ∼ 30–60% compared to the actual Uchuu halo pairwise velocities. This is a limitation of the Eisenstein-Hu power spectrum approximation; using CAMB or CLASS, or calibrating the power spectrum from the simulation, would reduce this bias.

5.3 BBC makes P23 worse, not better

C25 found that BBC *increases* the P23 bias from −20% (simple fit) to −26% (BBC fit). We reproduce this on our SNANA simulations (Fig. 6c): with 2125 SALT3-fitted low-z P23 SNe, the Gaussian bias goes from −98% (no BBC, boundary hit) to −28% (with BBC). But the Student-t without BBC gives +14%, outperforming Gaussian + BBC. Adding BBC to the Student-t *worsens* the result to +25%.

Fig. 6a shows why: BBC barely changes the residual distribution. The excess kurtosis drops from 2.9 to 3.0 (effectively unchanged). BBC operates on the *mean* bias per (z, c, x_1) bin, but the $f\sigma_8$ bias comes from the *shape* of the distribution — the heavy tails that BBC cannot remove.

Fig. 6b shows the mechanism by which the Student-t succeeds where BBC fails. In the Gaussian likelihood, the top 10% of outlier SNe (sorted by standardized residual $|z_i|$) contribute 66% of the total χ^2. These are the P23 dust-contaminated SNe whose biased distances dominate the fit.

In the Student-t ($\nu = 6$), the same outliers contribute only 59% — still significant, but no longer dominant. This 7-point rebalancing is sufficient to shift $f\sigma_8$ from -98% to $+14\%$ bias.

The Student-t avoids BBC's fundamental limitation because it modifies the *likelihood*, not the *data*. The velocity vector $\boldsymbol{v}$ is untouched; only the statistical weight of each observation changes.

5.4 Caveats and limitations

The Student-t handles kurtosis, not skewness. The multivariate Student-t is symmetric: it has heavy tails but zero skewness. The P23 residuals have *both* skewness ($\gamma \approx 1$) and kurtosis ($\kappa \approx$ 5–8). The Student-t works because the $f\sigma_8$ likelihood is based on $\chi^2 = \boldsymbol{v}^T \boldsymbol{C}^{-1} \boldsymbol{v}$, a sum of *squared* terms that is insensitive to the sign of residuals. What biases $f\sigma_8$ is the excess of large-amplitude residuals (kurtosis), not their asymmetry (skewness). The skewness shifts the mean $\Delta\mu$, but this is absorbed by the Tripp M_0 parameter.

M_0 is not in the likelihood. The Tripp fit (step 1: M_0, α, β) and the $f\sigma_8$ fit (step 2: velocity covariance) are sequential, not joint. M_0 absorbs the *global* mean of the P23 bias, but it is a single number. If the P23 bias varies with redshift — as it does, because the dust parameter τ is z-dependent — M_0 cannot absorb the z-dependent component. This residual creates a spurious radial velocity gradient that mimics $f\sigma_8$ signal.

Low-z vs. wide-z surveys. For our low-z PV survey ($z < 0.1$), τ varies by 25% across the sample, and the z-dependent bias is small enough that the Student-t alone suffices. For surveys spanning $z = 0.01$ to $z \sim 1$, the z-dependent mean shift would be large and the Student-t alone would not correct it. In that regime, both the Student-t (for the shape) and a z-dependent correction such as BBC or z-binned Tripp fitting (for the mean) would be needed. This is the correct complementary role for BBC: correcting the z-dependent mean bias that M_0 cannot absorb, while the Student-t handles the non-Gaussian shape that BBC cannot fix.

Simplified mocks. Our SNANA test uses 2125 SALT3-fitted SNe (partial results; the full 81k-SN run is ongoing). The cosmological prior is centered on the true $f\sigma_8$; in a blind analysis it would come from external data.

5.5 Beyond the Student-t: skew-t and forward models

The symmetric Student-t is a pragmatic first step, not the optimal solution. Two natural extensions would capture the skewness that the Student-t ignores:

Multivariate skew-t. The skew-t distribution (Azzalini & Capitanio 2003) extends the Student-t with a skewness vector $\boldsymbol{\delta}$ that allows asymmetric tails. The P23 skewness parameter could be estimated from the data alongside ν. The challenge is computational: the multivariate skew-t density involves the CDF of a $(N+1)$-variate t-distribution, which is expensive for $N \sim 6000$ SNe.

Forward model. A more principled approach would build the P23 dust model directly into the likelihood. For each SN with observed color c and host mass M, one would marginalize over the latent decomposition $c = c_{\rm int} + E_{\rm dust}$ using the known P23 priors:

$$p(v_i \mid f\sigma_8) = \int p(v_i \mid E_{\rm dust}, f\sigma_8)\, p(E_{\rm dust} \mid c_i, M_i)\, dE_{\rm dust}. \tag{8}$$

This would correctly model both the skewness and kurtosis for each SN, and would naturally produce z-dependent corrections because the dust priors are z-dependent. The challenge is that the per-SN marginalization breaks the multivariate structure of the velocity covariance, requiring either approximations or MCMC sampling.

Both extensions are computationally more demanding but would address the residual +8–14% bias that the symmetric Student-t leaves. We leave their implementation to future work and note that the Student-t already recovers $f\sigma_8$ to within 1σ of the fiducial for all Uchuu mocks tested.

6 CONCLUSIONS

We have shown that replacing the Gaussian velocity likelihood with a multivariate Student-t distribution effectively corrects the $f\sigma_8$ bias from P23 non-Gaussian intrinsic scatter:

(i) On Uchuu N-body mocks with 6600 SNe and P23 scatter, the bias is reduced from -41% (Gaussian) to $+8\%$ (Student-t), with the residual from the analytical covariance model.

(ii) The method is fully data-driven: ν is estimated from the kurtosis, requiring no knowledge of the scatter model.

(iii) The only modification is one term in the log-likelihood (Eq. 5 vs. Eq. 1).

(iv) For near-Gaussian scatter models, the method automatically recovers the standard Gaussian likelihood.

As SN Ia samples grow to $\mathcal{O}(10^5)$ with LSST, the Student-t likelihood provides a practical solution for peculiar velocity analyses with realistic dust-based scatter models.

ACKNOWLEDGEMENTS

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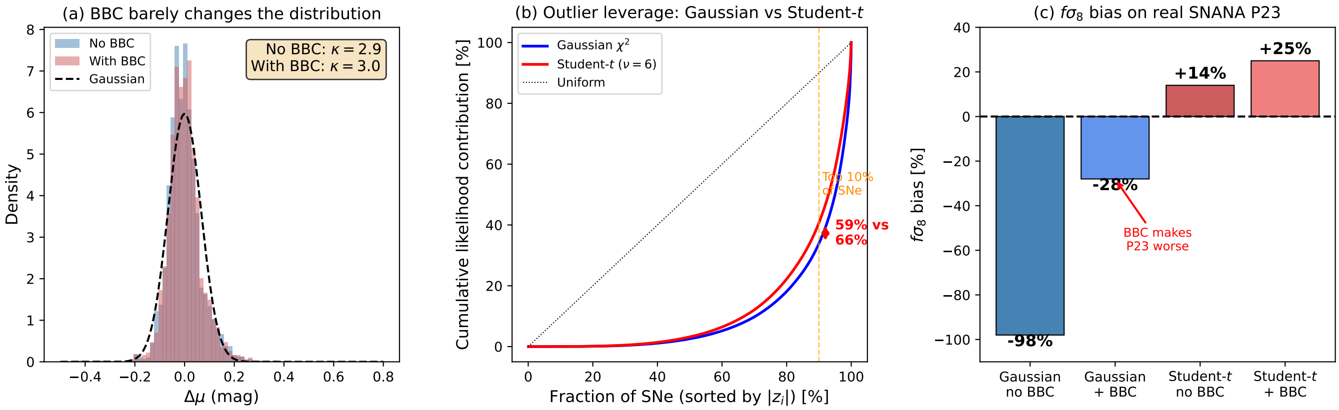

Figure 6. **(a)** BBC barely changes the Hubble residual distribution: kurtosis is ~ 3 with or without BBC. **(b)** Cumulative likelihood contribution sorted by residual size. In the Gaussian (blue), the top 10% of outlier SNe contribute 66% of χ^2. The Student-t (red) reduces this to 59%, breaking the outlier dominance. **(c)** $f\sigma_8$ bias on real SNANA P23 data (2125 SNe): the Student-t without BBC (+14%) outperforms Gaussian + BBC (−28%). BBC worsens both likelihoods.